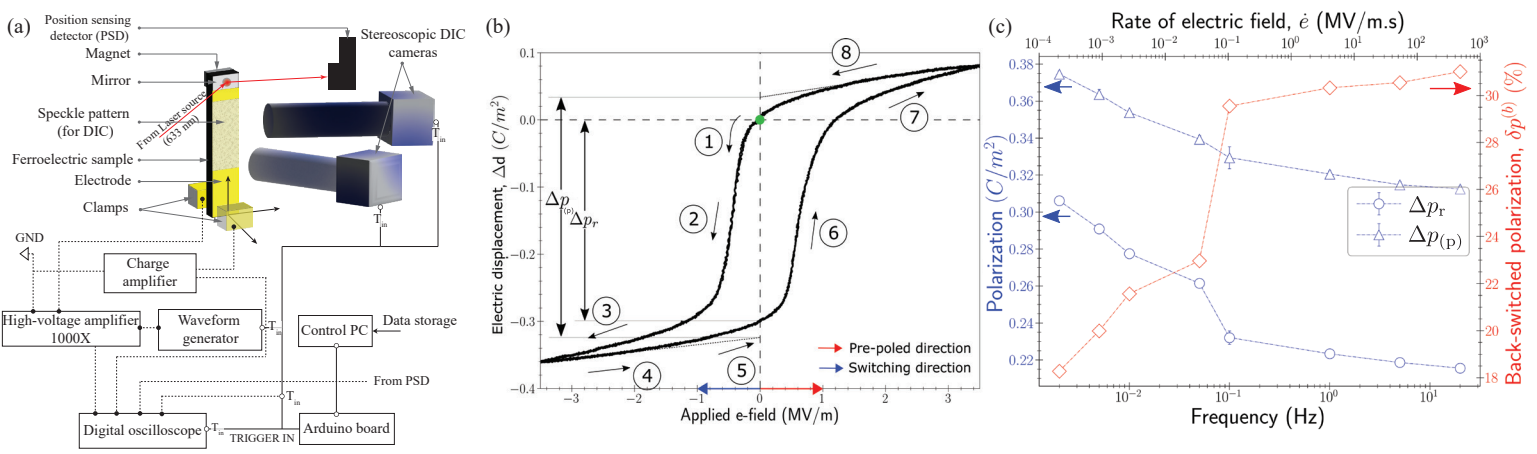

(a)
Position sensing detector (PSD)
Magnet
Mirror
Speckle pattern (for DIC)
From Laser source (633 nm)
Ferroelectric sample
Electrode
Clamps
Stereoscopic DIC cameras
GND
Charge amplifier
High-voltage amplifier 1000X
Waveform generator
Control PC
Data storage
From PSD
Digital oscilloscope
TRIGGER IN
Arduino board
(b)
Electric displacement, Δd (C/m²)
Applied e-field (MV/m)
Pre-poled direction
Switching direction
(c)
Rate of electric field, ė (MV/m.s)
Polarization (C/m²)
Back-switched polarization, δp(b) (%)
Frequency (Hz)